# Legislative Coordination and Institutional Change

A Bipartite Network Analysis of Coalition Dynamics in the Brazilian Chamber of Deputies

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Slides & paper

# The puzzle

- Polarization should make legislatures **ungovernable**.
- Brazil 2014–16: impeachment · corruption scandals · recession · polarization.
- Yet the Chamber **kept forming majorities**, across impeachment and a change of power.
- The literature's crisis typology: **collapse · fragmentation · realignment · institutionalization**. Brazil fits **none**.
- **How? And at what cost?**

# Coalitional presidentialism, and a cluster of shocks

- Strong president, **no majority** → govern by **coalition** (Abranches 1988).
- Levers: **budget · posts · agenda**; the **Centrão** sells support as the president's *agent*.
- Recent work reads a **weakening presidency and a rising Congress**, much of it via **budget powers** once a pillar of the president's toolkit.
- 2014–16, all at once: impeachment · corruption scandals · recession · **and the president's budget lever removed**.
- **Shocks co-occur in a tight 2014–16 window** → no single lever separates cleanly from the rest (hence the eliminative design).

# The structural question, and some rival explanations

- The open question is structural: **How did coordination reorganize after the 2014–2016 shocks? What happened to the pragmatic center, and at what cost?**
- **Then rule out the rival explanations, one at a time:**
- Is it **turnover** (new members) or a one-off **crystallization**?
- Is it the **Cunha effect** (the agenda-setter) or the **fiscal channel**: does re-sorting concentrate on economic-fiscal rather than rights-and-moral bills?

*The tests ask whether a durable pivot remains once turnover, agenda, and fiscal explanations are removed.*

# Why networks, not ideal points

- **deputies × bills** → agreement graph → **communities**.
- Standard **one-dimensional ideal-point** models compress change onto one axis.
- Blocs **recompose** while ideology looks stable; an axis can't see it.
- No axial assumption → the shift is **bloc recomposition**.

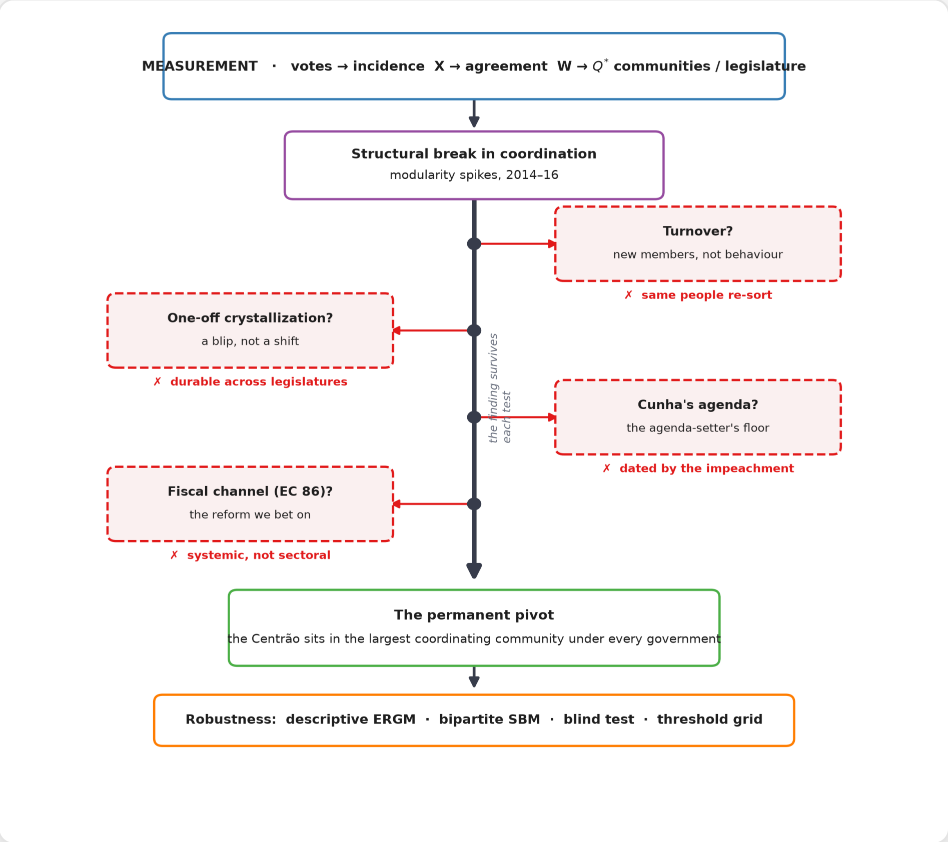
# Related work, and how this differs

- **Prior networks of the Chamber exist** (Brito 2020; Toledo Jr. 2025). **Four differences:**
- **1. Bipartite primitive:** starts from the deputy×bill incidence and explicitly tests projection dependence.
- **2. Null-corrected structure:** modularity against a **within-vote permutation null (Q*)**, not a raw score.
- **3. Within-legislator design:** separates a change in behavior from a change in composition.
- **4. Theory-driven object:** the object is **which coordinating community a bloc occupies relative to the executive**, not a left–right axis.

# The methodological approach

- **Measure:** roll-calls → **deputy × bill** graph → **communities per legislature**.
- **Classify bills:** fiscal vs moral, **rule-based** + local **LLM cross-check** (Qwen 2.5 32B), human-validated (n=294, α=0.84, macro-F1 0.88, 89.5%).
- **Compare:** across the 2014–16 break; a **within-deputy panel** separates behavior from turnover.
- **Falsify, not estimate:** near-collinear shocks → **eliminate rivals one at a time**.
- **Stress-test:** Q* null, 12-cell threshold grid, alternative projections and algorithms.

# The design at a glance




One finding survives while each rival explanation is ruled out in turn.

# Votes as a bipartite incidence matrix

$$X_{ij} = \begin{cases} +1 & \text{deputy } i \text{ voted } \textbf{Yes} \\ -1 & \text{voted } \textbf{No} \\ 0 & \text{abstention, obstruction, or absent} \end{cases}$$

Deputy × bill, before projecting to the agreement graph.

# The coordination metric: agreement score

$$W_{ij} = \frac{|\{v : S_{iv} = S_{jv} \neq 0\}|}{|\{v : S_{iv} \neq 0 \text{ and } S_{jv} \neq 0\}|}, \qquad S = \mathrm{sign}(X)$$

**where** $W_{ij}$ = agreement rate between deputies $i$ and $j$ • $S = \mathrm{sign}(X)$ = signed vote matrix ($+1$ Yes, $-1$ No, $0$ otherwise) • $v$ = a recorded vote

numerator = votes where $i, j$ agree • denominator = votes both attended • edge set to $0$ if fewer than $30$ shared votes

Krehbiel (1993) co-voting agreement, normalised by joint attendance.

# The ERGM specifications I tried

$$\Pr(Y = y) = \kappa^{-1} \exp\Big( \sum_k \theta_k \, z_k(y) \Big)$$

1. **bipartite pre-test** $z:\ \text{edges} + \text{gwb1deg} + \text{gwb2deg}$ **DID NOT MIX**
2. **cross-sectional** $z:\ \text{edges} + \text{nodematch(party, Centrao, gov/opp, UF)}$ **RUNS**
3. **temporal (btergm)** $z_t:\ \text{edges} + \text{gwesp} + \text{homophilies} + \text{memory}$ **RUNS**

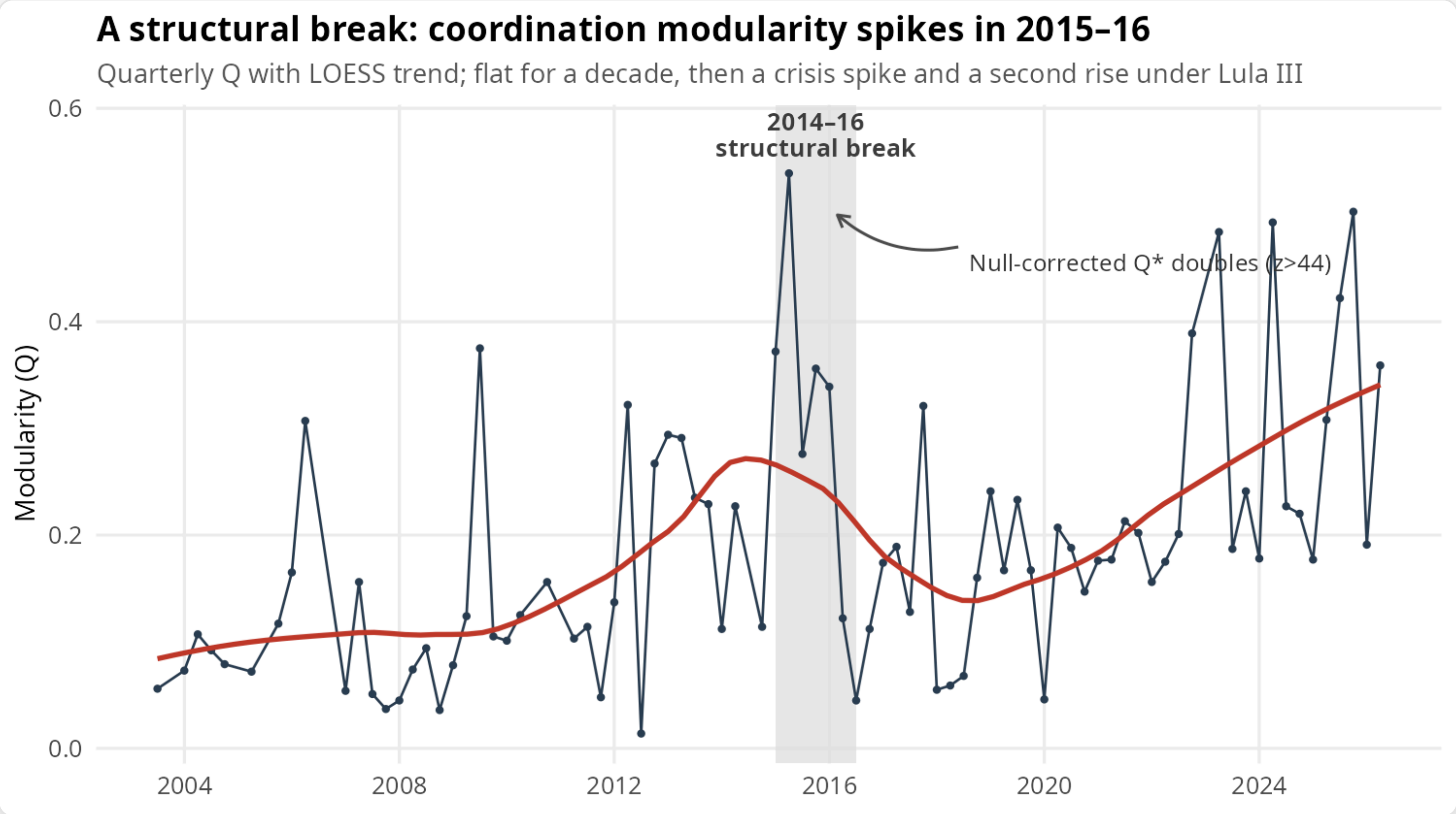
**statistics** $\text{edges} = \sum_{i<j} y_{ij}$ • $\text{nodematch}_x = \sum_{i<j} y_{ij}\, \mathbf{1}[x_i = x_j]$ • gwesp = geom.-weighted shared partners (closure) • memory = tie persistence across $t$

Same estimand throughout, the marginal effect of EC 86; the binding constraint is identification, not mechanics.

# What the ERGM exercises establish

- Estimand: the **causal effect of mandatory amendments** (EC 86).
- **Static bipartite ERGM** (deputy×bill incidence): **did not mix** on the dense 54L network.
- **Temporal agreement-network model** (btergm): **runs**, recovering **closure** and **tie persistence**.
- **Identification remains:** neither exercise separates EC 86 from the post-2014 shock cluster (VIF 11.6).
- **Implication:** estimation can *model* coordination, but cannot *isolate* the reform's marginal effect.

# A structural break: modularity spikes in 2015–16




Quarterly modularity rises from ≈.15 to .51 in 2016.

# Null-corrected modularity, Q*

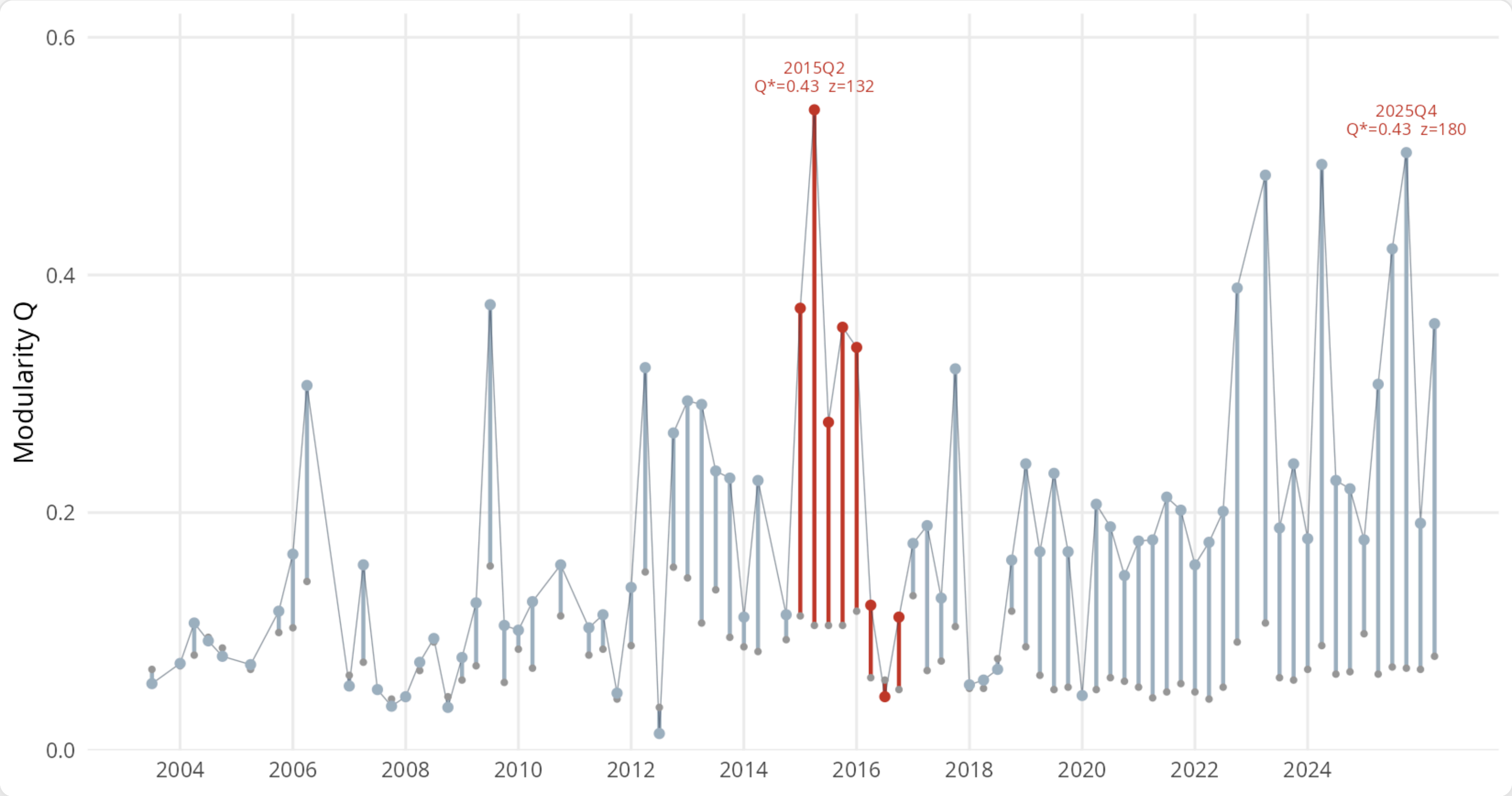
$$Q = \frac{1}{2m} \sum_{ij} \left( A_{ij} - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j)$$

$$Q^* = Q_{\text{obs}} - \langle Q_{\text{null}} \rangle, \qquad z = \frac{Q_{\text{obs}} - \mu_{\text{null}}}{\sigma_{\text{null}}}$$

**where** $A_{ij}$ = agreement-graph weight • $k_i$ = weighted degree of $i$ • $m$ = total edge weight • $\delta(c_i, c_j) = 1$ if $i, j$ share a community, else $0$ • null = within-vote permutations

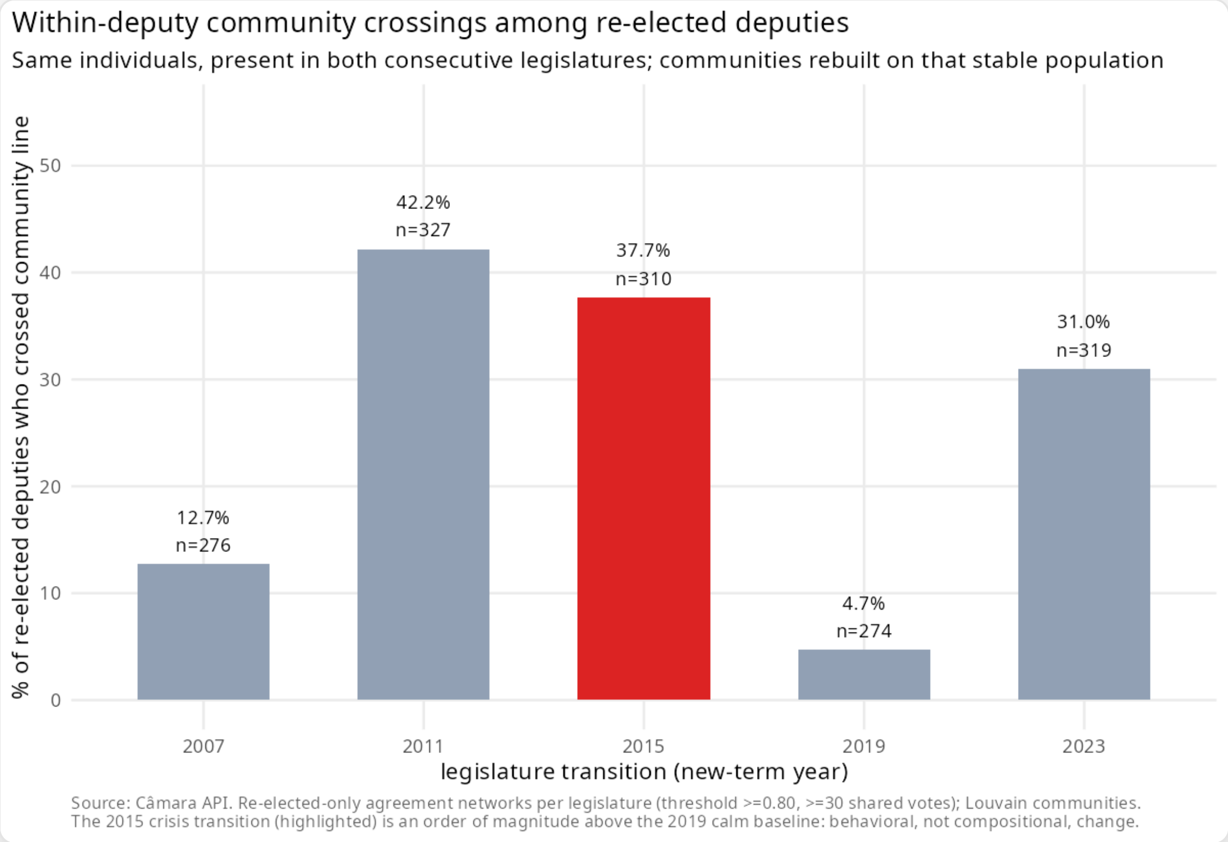
Newman–Girvan modularity, benchmarked against a permutation null: Q* is what exceeds chance.

# Null-corrected modularity (Q*) over time



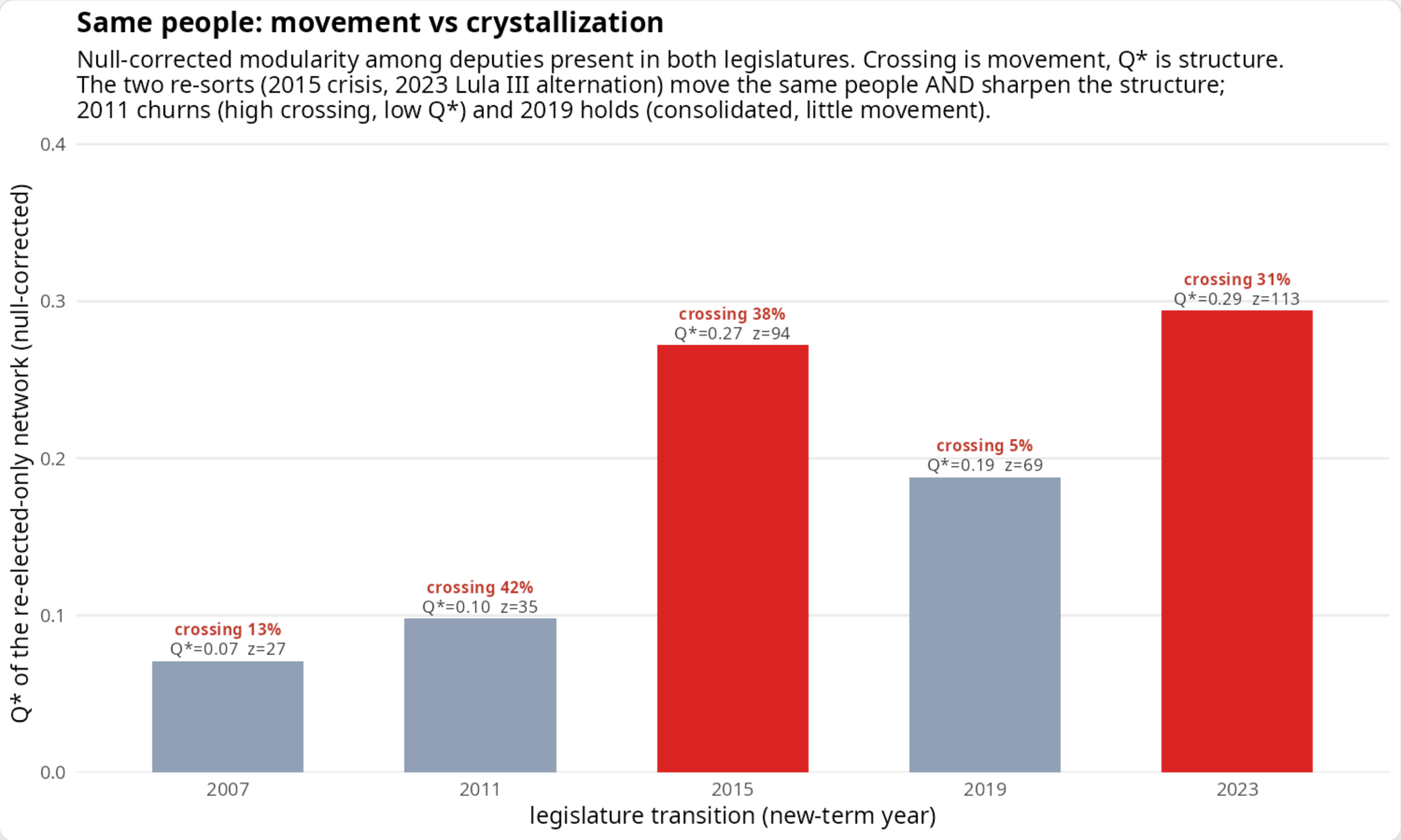

Observed Q exceeds the within-vote null, peaking in 2015–16 and rising again under Lula III.

# Same people, not turnover



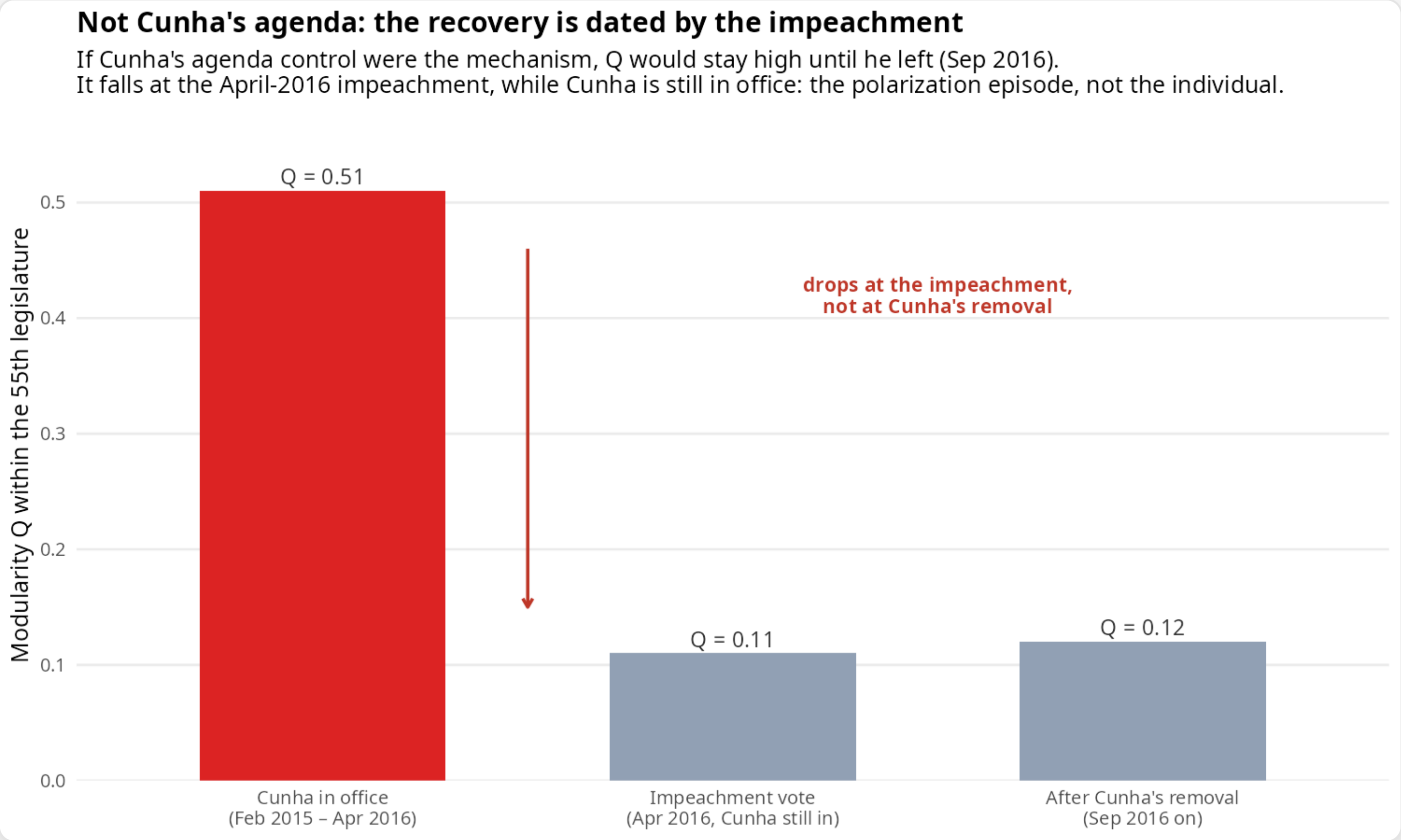

Reelected-deputy crossing: **37.7%** at the break, **4.7%** afterward.

# Movement is not crystallization



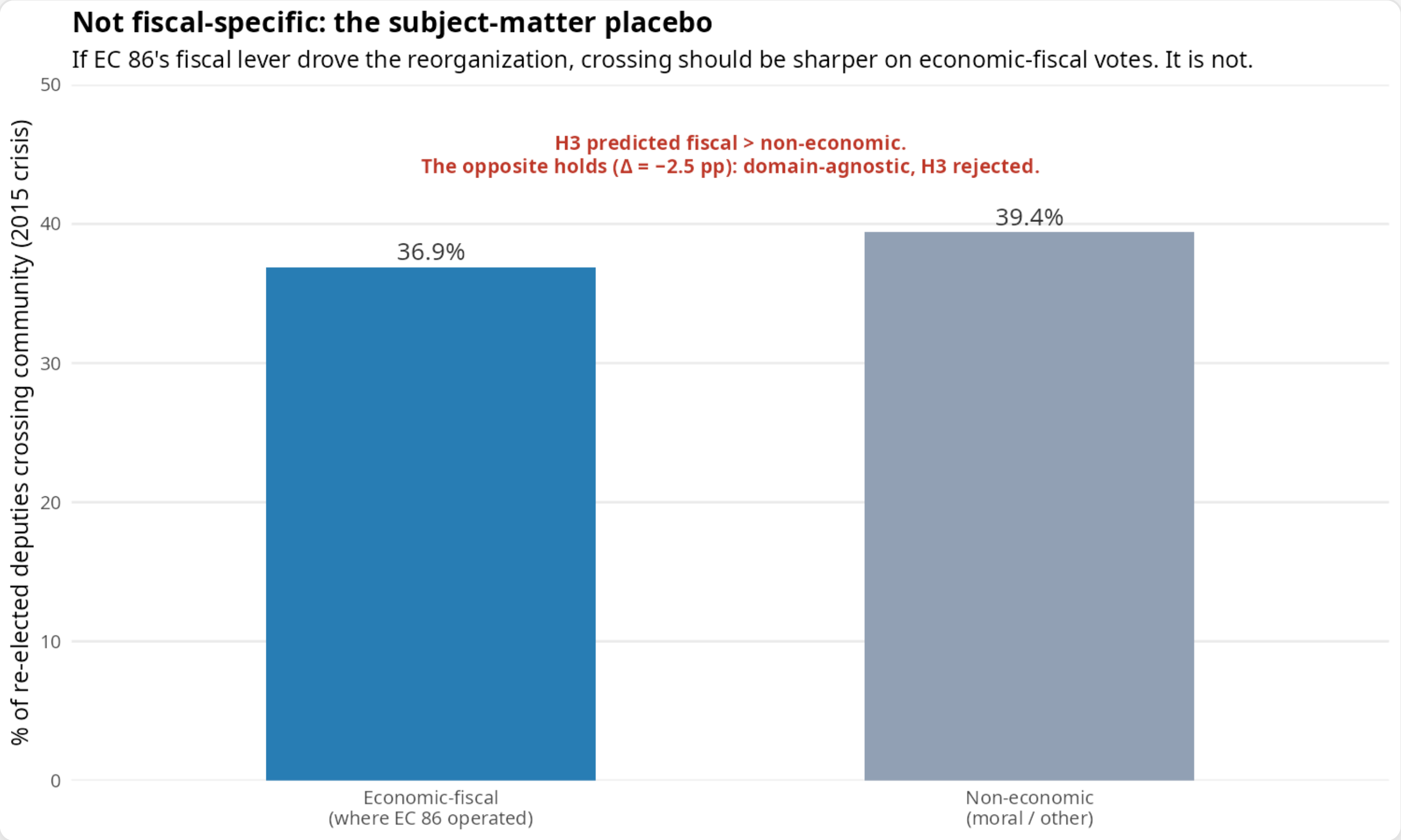

2011 moves more deputies but stays diffuse; 2015 and 2023 sharpen the structure.

# Not Cunha's agenda: dated by the impeachment



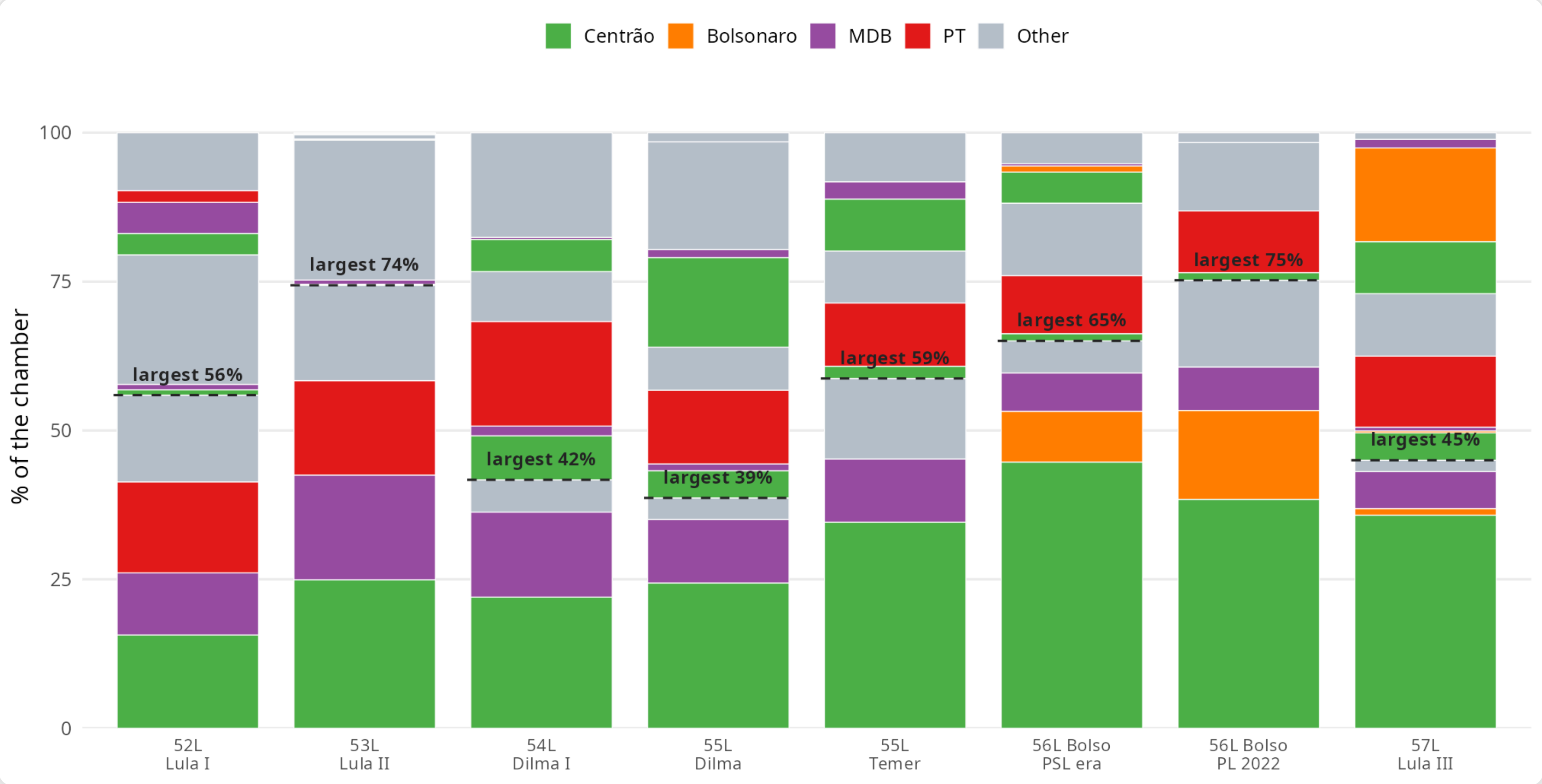

Modularity falls at the impeachment, not at Cunha's removal.

# Not fiscal-specific: the subject-matter placebo



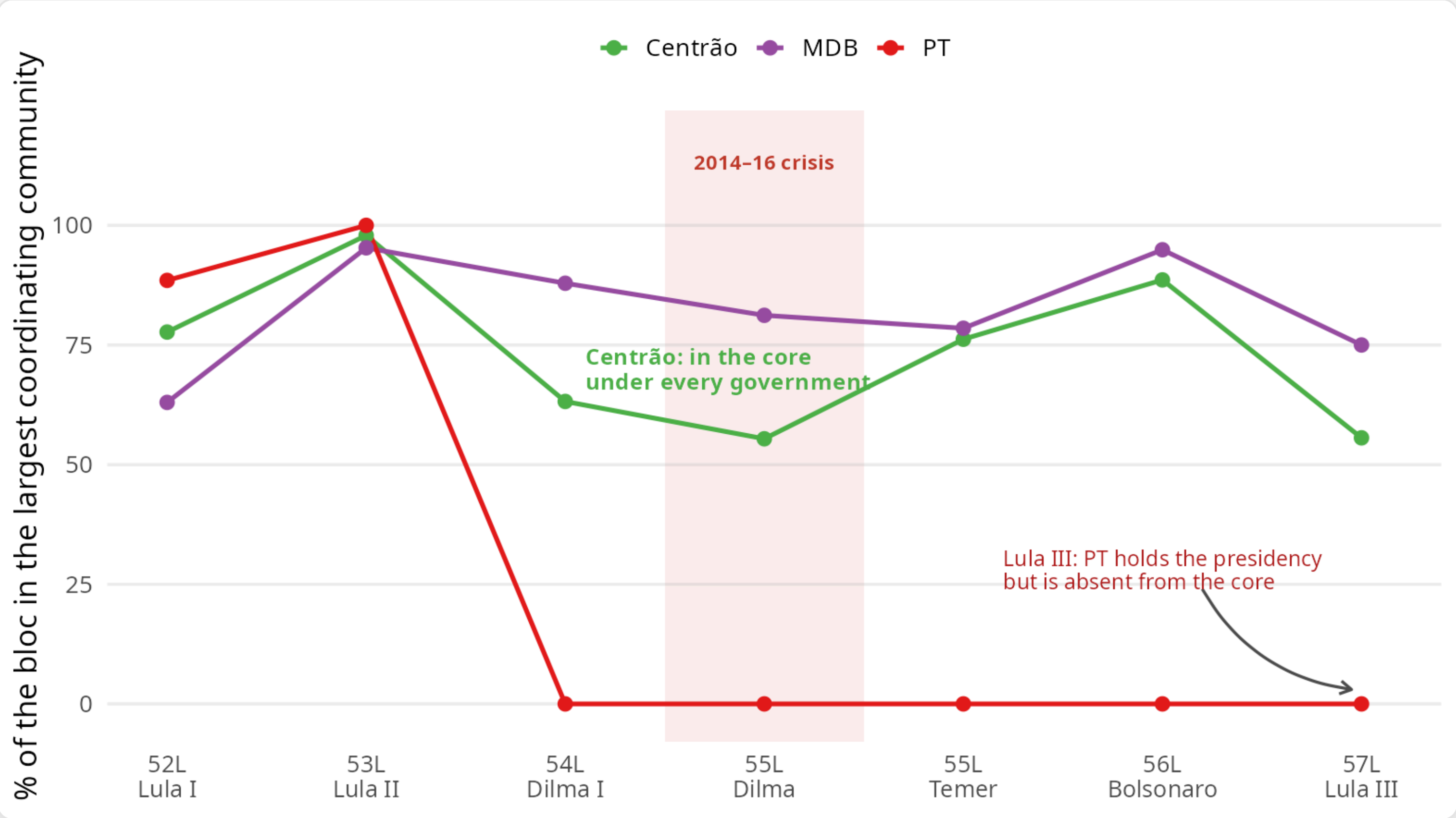

Fiscal: 36.9%; non-economic: 39.4%; Δ = −2.5 pp.

# The coordinating communities, sized




From 2014, the Centrão anchors the largest community while the PT moves to a smaller one.

# The pivot: in the largest community under every government

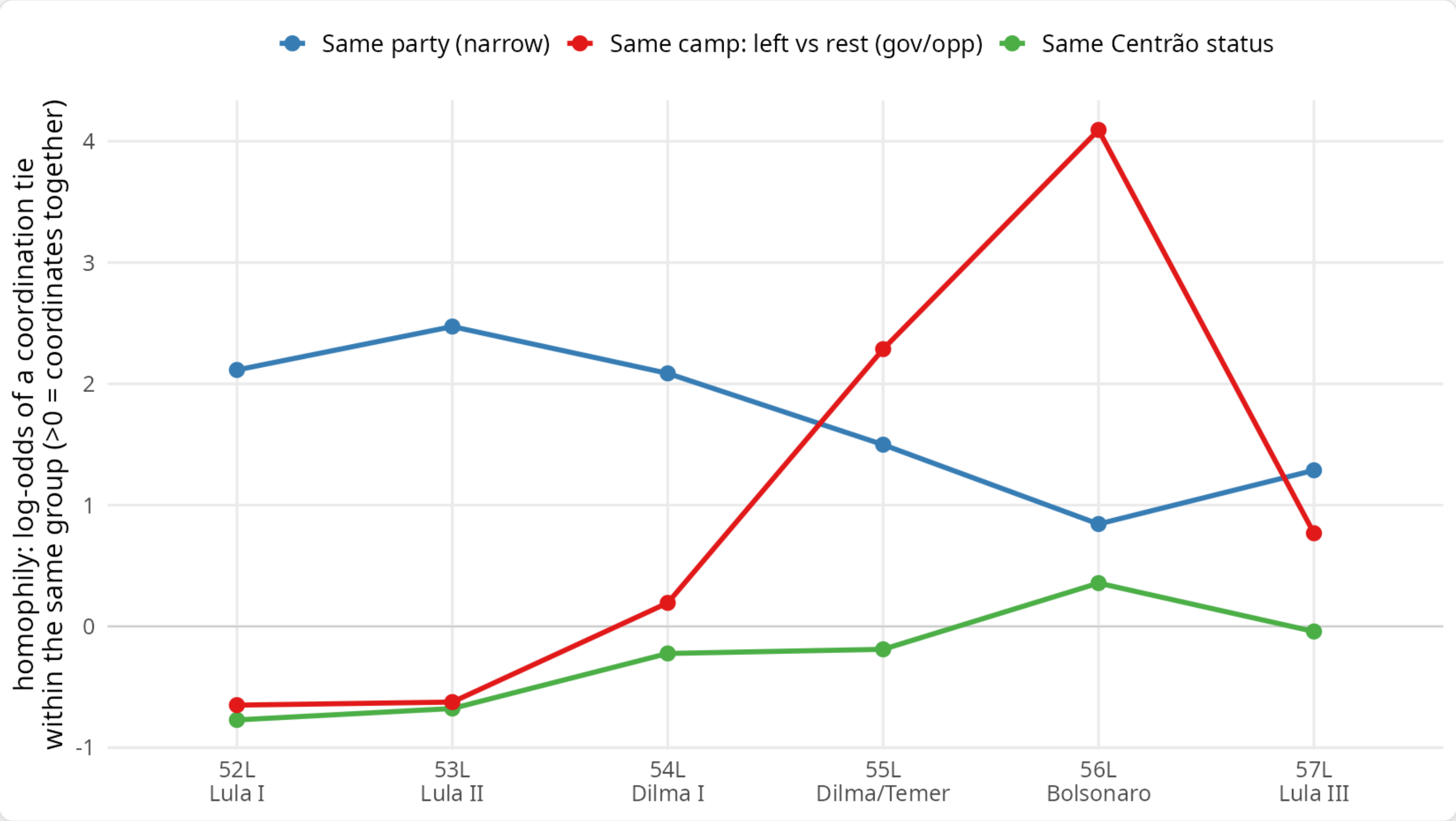



The Centrão remains in the largest community across governments; the PT does not.

# Can we trust these communities?

- **Noise?** $Q^*$ sits far above a within-vote permutation null ($z > 44$).
- **Projection?** A projection-free bipartite SBM shows compatible alignment at the 56L (ARI 0.23), weakening the projection-artifact interpretation.
- **Labels?** Vote-only communities increasingly align with an independently coded Centrão (ARI 0 → 0.39).
- **Model dependence?** A descriptive ERGM detects the same shift, from party homophily to government–opposition organization.
- **Just the PT in opposition?** No: the government side organizes as strongly as the opposition, and the shift survives dropping the PT.

# What organizes coordination: party gives way to position




Party homophily declines as government–opposition organization rises after 2014.

# The thesis, and what it means

- **The pragmatic center is now the recurring pivot governments build majorities through**, coordinating by **position relative to the executive**, not a fixed ideological axis.
- The Centrão is **old** (1987); what changed is its **position**, *agent* → *pivot*.
- **Why (a reading, not a tested cause):** the budget shifted to Congress (**mandatory amendments**, then the **secret budget** and **agenda control**), so the tool that once disciplined the pivot now lets it **set the terms** (*governo congressual*).
- **What it means:** governability **sold, not represented**; the cost is **accountability**, not gridlock.

# Contribution, and what the case teaches

- **Measurement:** recover the Centrão's permanence from the votes, its position and persistence, where the literature had narrative.
- **Design:** with near-collinear shocks, eliminate the rival explanations rather than estimate one effect.
- **A portable template:** turn a narrative prediction, that institutional crisis reshapes the legislature, into a measurable, falsifiable object, and narrow the viable mechanisms.
- **A way of looking, not a new claim:** the permanent-pivot type is old (Israel, post-war Italy).

# Next steps

- **Separate influence from selection:** a contagion-vs-similarity REM (Leifeld et al. 2026) or a SAOM; my bipartite data has the structure.
- **A formal inferential backbone:** a degree-corrected SBM, and a validated backbone (Neal's SDSM) beyond the edge filters.
- **Quantify evidence of absence:** Bayes factors on the null placebos.
- **Travel the type:** the permanent governing pivot elsewhere (Israel, post-war Italy).

*I'd value your pointers and collaborations on any of these.*

# Thank you

Questions welcome.

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